IBRAHIMA DIOUF, PH.D.

Lecturer | Climate and Health Researcher | International Consultant

Cheikh Anta Diop University (ESP/UCAD), Senegal

University of Labé (UL), Guinea

ibrahima23.diouf@ucad.edu.sn & ibrahima.diouf@univ-labe.edu.gn

https://scientific-hub-lab.github.io/Dr-Ibrahima-Diouf/

PROFILE

- Over 15 years of interdisciplinary experience in climate and health research, with strong expertise in climate variability, vector-borne disease modeling, and health system resilience.
- Ph.D. in Climate and Health Impacts with a focus on seasonal forecasting, climate modeling, and public health risks in West Africa.
- Extensive experience in national and international projects funded by organizations such as the European Union (EU FP7), USAID, WHO, UNICEF, NOAA, and IRD.
- Proven track record in academic teaching, project coordination, scientific writing, and stakeholder engagement.
- Currently serving as Lead Author for Chapter 7 (Africa) of the IPCC Seventh Assessment Report (AR7), Working Group II.

AREAS OF EXPERTISE

- **Climate and Health:**
  - Specialized in the study of climate-health interactions, with a focus on seasonal forecasting in West Africa and the impact of climate variability and change on vector-borne diseases, particularly malaria.
- **Research and Capacity Building:**
  - Researcher at Cheikh Anta Diop University of Dakar and lecturer-researcher at the University of Labé in Guinea Conakry, actively involved in strengthening national capacities to address climate and health challenges, in collaboration with the Ministry of Health in Senegal.
- **Consultancy and International Projects:**
- Consultant for projects funded by UNICEF and USAID, focusing on health system resilience. Extensive experience enhanced by a collaboration with NOAA (National Oceanic and Atmospheric Administration) in the United States since 2017.
- **Training and Supervision:**
  - Trainer and thesis supervisor at several universities in Senegal and Guinea, contributing to the education and mentoring of future generations of researchers.

EDUCATION

- **École Polytechnique Supérieure, Cheikh Anta Diop University of Dakar (ESP/UCAD)**
  Ph.D. in Climate and Health Impacts, January 2010 - November 2016
- **École Polytechnique Supérieure, Cheikh Anta Diop University of Dakar (ESP/UCAD)**
  Master's Degree in Meteorology, Oceanography, and Arid Zones Management, January 2009 - December 2010
- **Faculty of Science and Technology, Cheikh Anta Diop University of Dakar (FST/UCAD)**
  Master's Degree in Physics and Chemistry, January 2006 - December 2008

PROFESSIONAL AFFILIATIONS

- Francophone Team Lead - HMST - Global Fund (Climate-Health Integration in Senegal, Niger, Central African Republic and Guinea Bissau)
- Lead Author - IPCC AR7 (Chapter Africa)
- Expert Project: Delivering Inclusive Indicators for the Global Goal on Adaptation
  Organisation: International Institute for Environment & Development (IIED)
- Co-chair of the Applied Malaria Modeling Network (AMMnet, Senegal)
- Member of the American Meteorological Society (AMS)
- Member of the African Meteorological Society (AfMS)
- Member of the Standards Team for Official Statistics on Climate-Health Interactions
- Member of the International Expert Centre on Climate Change and Health (IECCCH)
- Contributing Author to the Thematic Document on Impacts and Adaptation for the International CMIP Project Office (CMIP-PO)

ACADEMIC, RESEARCH AND PROFESSIONAL EXPERIENCES

- **Lecturer-Researcher**, University of Labé, Guinea | Since Oct 2024
  Teaching and research activities focused on climate and health. Supervision of Master's and PhD students. Coordination of academic modules on environmental sciences.
- **Climate Expert**, Institut Pasteur de Dakar - WHO | Since May 2023
  Project on integrated surveillance and control of vector-borne diseases linked to climate change. Responsibilities include climate data analysis, contribution to risk mapping, and policy recommendations.
- **Consultant**, Ministry of Health and Social Action (MSAS) & Save the Children - UNICEF | Jun 2023-Aug 2024
  Involved in strengthening health system resilience in the face of climate-related risks. Led climate vulnerability assessments and supported strategy design.

- **Expert Climatique**, Centre de Suivi Écologique (CSE), Senegal | Aug 2023-Jan 2024
  Provided technical expertise on climate-health links. Conducted assessments on community vulnerability and contributed to adaptation policy planning.
- **Postdoctoral Researcher**, Ministry of Environment - NAP-GEF | Jul 2021-Dec 2023
  Contributed to Senegal's National Adaptation Plan. Designed and led training for stakeholders. Developed educational materials and advised ministries on adaptation data use.
- **Consultant & Trainer**, African Population and Health Research Center (APHRC) - GPSDD | May-Aug 2023
  Designed and delivered training on sustainable development data for health professionals. Strengthened national data analysis capacities.
- **Postdoc & Associate Scientist**, NOAA CPC / UCAR, USA | Jul 2017-Jul 2021
  Led modeling activities on malaria and climate variability in Africa. Developed early warning tools. Coordinated international collaborations on climate-health simulations.
- **Researcher**, PREFACE - Universidad Complutense de Madrid (UCM), Spain | Nov 2013-Oct 2017
  Studied the tropical Atlantic's influence on public health in West Africa. Integrated climate indicators into health policies.
- **Postdoctoral Researcher**, Centre de Suivi Écologique (CSE), Senegal | Sep 2016-May 2017
  Analyzed impacts of climate change on vector-borne diseases in West Africa. Supported the creation of policy tools for adaptation.
- **PhD Researcher**, Universidad Complutense de Madrid (UCM), Spain | Oct 2012-Jul 2015
  Research under Erasmus Mundus: Investigated oceanic and climate determinants of seasonal malaria dynamics in the Sahel.
- **Researcher**, UCM - UCAD Collaboration | May 2013-Oct 2014
  Updated resources for the Laboratory of Atmospheric and Ocean Physics (LPAO-SF). Contributed to cooperative research and student mentoring.
- **PhD Researcher**, University of Liverpool - QWeCI | Jan 2010-Jul 2013
  Built malaria-climate databases and tested dynamic disease models. Integrated seasonal forecasts into public health systems across West Africa.
- **Researcher**, Africa Adaptation Programme (AAP) - INTAC | Jan 2010-Jul 2012
  Supported the integration of climate adaptation into national planning. Worked with multidisciplinary teams and government stakeholders.
- **Researcher**, CORDEX-Africa - SSEESS, WCRP, SMHI | Jan 2016-Jul 2022
  Analyzed downscaled climate projections and modeled malaria responses in West Africa. Contributed to peer-reviewed publications and intercontinental workshops.

SELECTED PUBLICATIONS

- Maraun, D., Bojovic, D., Parker, W., Shepherd, T. G., Sobolowski, S., Caminade, C., Dosio, A., Gutowski, W. J., Sörensson, A., Diaz, L. B., **Diouf**, I., Doblas-Reyes, F. J., Hernández, V., Jack, C., Pereira, L., Rivera, J. A., Robledo, F., & Somot, S. (2026). The importance of accounting for stakeholder values, power relationships and language in constructing

relevant and trustworthy climate information. Earth's Future, 14(1), e2024EF005618. https://doi.org/10.1029/2024EF005618

- **Diouf**, I., Fall, P., Faye, A., Diouf, S., Diouf, A. K., Barry, M. B., ... & Sall, A. (2025). Assessment of Spatio-Temporal Trends in Rainfall Indices in Senegal: Validation of CMIP6 Models over the Historical Period and Projections Under Future Climate Scenarios. Climate, 13(12), 247.
- Ruane, A. C., Pascoe, C. L., Teichmann, C., Brayshaw, D. J., Buontempo, C., **Diouf, I.**, Fernández, J., Gonzalez, P. L. M., Hassler, B., Hernaman, V., Im, U., Iovino, D., Juckes, M., Lake, I. L., Lam, T., Lin, X., Mao, J., Nazarian, N., Parey, S., Roy, I., Tseng, W.-L., Turner, B., Wiebe, A., Zhao, L., & Zurell, D. (2024). CMIP7 data request: Impacts and adaptation priorities and opportunities. Geoscientific Model Development, 18(23), 9497-9540. https://doi.org/10.5194/gmd-18-9497-2024
- Goudiaby, O., Bodian, A., Dezetter, A., Ogilvie, A., **Diouf, I.**, & Ndiaye, P. M. (2025). Hydrological evaluation of gridded rainfall products for streamflow simulation in West Africa. Journal of Hydrology: Regional Studies, 62, 102984.
- Talla, C., Diarra, M., **Diouf, I.**, Thiam, M. S., Gaye, A., Barry, M. A., ... & Loucoubar, C. (2025). Impact of climatic factors on malaria in Senegal based on the surveillance system between 2015 and 2022. Frontiers in Tropical Diseases, 6, 1631996.
- Toure, M., Sy, **I., Diouf, I.**; Gueye, O.; Bekele, E., Bhuiyan, M.A.E., Sambou, M.J., Ndiaye, P.N., Thiaw, W.M., Badiane, D.; Diongue-Niang, A., Gaye, A.T., Ndiaye, O., Faye, A. Machine Learning-Based Prediction of Heatwave-Related Hospitalizations: A Case Study in Matam, Senegal. Int. J. Environ. Res. Public Health 2025, 22, 1349.
- Touré M., Sy I., **Diouf I.**, Gueye O., Bekele E., Bhuiyan M.A.E., Sambou M.J., Ndiaye P.N., Thiaw W.M., Badiane D., Niang A.D., Gaye A.T., Ndiaye O., Faye A. *Machine Learning-Based Prediction of Heatwave-Related Hospitalizations: A Case Study in Matam, Senegal.* International Journal of Environmental Research and Public Health, accepté le 16 juin 2025.
- **Diouf I.**, Fall P., Faye A., Diouf S., Diouf A.K., Barry M.B., Bodian A., Sall A. *Assessment of Spatio-Temporal Trends in Rainfall Indices in Senegal: Validation of CMIP6 Models over the Historical Period and Projections under Future Climate Scenarios.* Climate, accepté le 16 juin 2025.
- Diouf D., Martín-Rey M., Rodríguez-Fonseca B., **Diouf I.**, Dione C., Akinbobola A., Gaye A.T. *Influence of Environmental Variability on Meningitis in West African Countries: Pre- and Post-Vaccination.* Frontiers in Environmental Health, 2025, 4:1531076. https://doi.org/10.3389/fenvh.2025.1531076
- **Diouf I.**, Fall P., Diakhaté M., Senghor H., Dieng M.D.B., Tompkins A.M., Arnault J., Sié A., Fersch B., Laux P., Schwarz M., Zabré P., Munga S., Khagayi S., **Diouf I.**, Kunstmann H. (2024). *Process-based atmosphere hydrology-malaria modeling: Performance for spatio-temporal malaria transmission dynamics in Sub-Saharan Africa.* Water Resources Research, 60(6), e2023WR034975.
- Goudiaby O., Bodian A., Dezetter A., **Diouf I.**, Ogilvie A. (2024). *Evaluation of Gridded Rainfall Products in Three West African Basins.* Hydrology, 11(6), 75.
- **Diouf I.**, Sy I., Ndione J.-A, Gaye A.T. (2024). *An overview of Climate Change Impacts on the Health Sector for the Research Program of the National Adaptation Plan Support Project in Senegal.* J. P. Soaphys, http://dx.doi.org/10.46411/jpsoaphys.2023.016 Vol 3, N°2 (2023) C23N22

- **Diouf I.**, Sy I., Diakhate M. (2024). *Assessing Climate Change Impacts on Public Health in Haiti: A Comprehensive Study of Disease Distribution, Modeling, and Adaptation Strategies.* Frontiers in Tropical Diseases, 4, 1287499.
- Fall P., **Diouf I.**, Deme A., Diouf S., Sene D., Sultan B., Janicot S. (2023). *Enhancing Understanding of the Impact of Climate Change on Malaria in West Africa Using VECTRI and Bias-Corrected CMIP6 Data.* Microbiology Research, 14(4), 2148-2180.
- Fall P., **Diouf I.**, Deme A., Diouf S., Sene D., Sultan B., Famien A.M., Janicot S. (2023). *Bias-Corrected CMIP5 Projections for Climate Change and Assessments of Impact on Malaria in Senegal under the VECTRI Model.* Tropical Medicine and Infectious Disease, 8(6), 310. https://www.mdpi.com/2414-6366/8/6/310
- **Diouf I.**, Ndione J.-A, Gaye A.T. (2022). *Malaria in Senegal: Recent and Future Changes Based on Bias-Corrected CMIP6 Simulations.* Tropical Medicine and Infectious Disease, 7(11), 345.
- Fall P., **Diouf I.**, Deme A., Sene D. (2022). *Assessment of Climate-Driven Variations in Malaria Transmission in Senegal Using the VECTRI Model.* Atmosphere, 13, 418. https://doi.org/10.3390/atmos13030418
- **Diouf I.**, Suárez-Moreno R., Rodríguez-Fonseca B., Caminade C., Wade M., Thiaw M.W., Deme A., Morse A.P., Ndione J.-A., Gaye A.T., Diaw A., Ndiaye M.K.N. (2021). *Oceanic Influence on Seasonal Malaria Incidence in West Africa.* Weather, Climate, and Society, 14. https://doi.org/10.1175/WCAS-D-20-0160.1
- **Diouf I.**, Gbenga J.A., Abiodun M.A., Christopher L., Joyce M.S., Pascal Y., Ndione J.-A., Emiola O.G. (2021). *Impact of Future Climate Change on Malaria in West Africa.* Theoretical and Applied Climatology, 1-13. https://doi.org/10.1007/s00704-021-03807-6
- Kante I., **Diouf I.**, Millimono T., Kourouma J. (2021). *COVID-19 in Conakry, Guinea: Analysis and Relationship with Meteorological Factors.* Atmospheric and Climate Sciences, 11. https://doi.org/10.4236/acs.2021.112018
- Faye M., Dème A., Diongue A.K., **Diouf I.** (2021). *Impact of Different Heat Wave Definitions on Daily Mortality in Bandafassi, Senegal.* PLoS ONE, 16(4):e0249199. https://doi.org/10.1371/journal.pone.0249199
- **Diouf I.**, Sy S., Senghor H., Fall P., Diouf D., Diakhate M., Thiaw M.W., Gaye A.T. (2021). *Potential Contribution of Climate Conditions on COVID-19 Transmission over West and North African Countries.* Atmosphere, 13(1). https://doi.org/10.3390/atmos13010034
- **Diouf I.**, Rodríguez-Fonseca B.R., Caminade C., Thiaw W.M., Deme A., Morse A.P., Ndione J.-A., Gaye A.T., Diaw A., Ndiaye M.K.N. (2020). *Climate Variability and Malaria over West Africa.* American Journal of Tropical Medicine and Hygiene, 102(5), 1037-1047. https://doi.org/10.4269/ajtmh.19-0062
- Kante I.K., Sall S.M., Badiane D., **Diouf I.**, Dieng A.L., Diaby I., Guichard F. (2020). *Analysis of Rainfall Dynamics in Conakry, Guinea.* Atmospheric and Climate Sciences, 10(1), 1. https://doi.org/10.4236/acs.2020.101001
- **Diouf I.**, Fonseca B.F., Deme A., Caminade C., Morse A.P., Cisse M., Sy I., Dia I., Ermert V., Ndione J.-A., Gaye A.T. (2017). *Comparison of Malaria Simulations Driven by Meteorological Observations and Reanalysis Products in Senegal.* International Journal of Environmental Research and Public Health, 14(10), 1119. https://doi.org/10.3390/ijerph14101119
- **Diouf I.**, Deme A., Ndione J.-A., Gaye A.T., Rodríguez-Fonseca B., Cissé M. (2013). *Climate and Health: Observation and Modeling of Malaria in the Ferlo (Senegal).* Comptes Rendus Biologies, 336(5-6), 253-260. https://doi.org/10.1016/j.crvi.2013.04.001

CONFERENCES / WORKSHOPS

▪ Developing a Shared African Climate Literacy (CORDEX-Africa Workshop 2)**,** University of Cape Town, Cape Town (South Africa), 24-26 March 2026

▪ Missions d'assistance technique Climat & Santé - HMST (Fonds mondial)**,** Afrique (multi-pays), 2026
Team Lead (Sénégal, Niger, République centrafricaine) & Membre de l'équipe (Burundi, Guinée-Bissau)

- Sénégal, 2-8 février 2026
- Burundi, 9-15 février 2026
- République centrafricaine, 16-22 février 2026
- Niger, 23-27 février 2026
- Guinée-Bissau, 13-18 mars 2026

- Expert Meeting on Climate Change and Health, Wellcome Trust, London (United Kingdom), 27-29 January 2026
- IPCC AR7 First Lead Author Meeting, Paris (France), 1-5 December 2025 Participation in the inaugural meeting bringing together 600+ IPCC experts to initiate the drafting of the Seventh Assessment Report (AR7).
- Réunion annuelle du CSID Network (*Climate Sensitive Infectious Disease Network*), KanchanaInvité officiel par Mahidol-Oxford Tropical Medicine Research Unit (MORU) Participation à des discussions stratégiques sur les maladies infectieuses sensibles au climatburi (Thaïlande), *juillet 2025.*
- Global Goal on Adaptation (GGA), SB62 UNFCCC, Bonn (Germany), June 2025
- Validation Workshop on Vulnerability Studies, Dakar (Senegal), May 2025 - Oral presentation on climate and health impacts
- FE2025 Malaria Modellers Capacity Building, Iba Der Thiam University, Thiès (Senegal), April 2025 - Trainer
- SOSEPIT 2nd Congress, Dakar (Senegal), April 2025 - Oral presentation on tropical diseases
- Pre-SB62 Strategic Meeting, Nairobi (Kenya), February 2025 - Participant with AGNES and partners
- Scientific Seminar, University of Labé (Guinea), December 2024 - Oral presentation on climate-health impacts
- Climate & Malaria Training, INES-Ruhengeri (Rwanda), December 2024 - Trainer with Expertise France
- MSc Seminar, UCM (Spain), November 2024 - Oral presentation on health impacts in Senegal
- 8th World One Health Congress, Cape Town (South Africa), September 2024 - Poster presentation
- Climate & Health Conference, Labé (Guinea), July 2023 - Keynote speaker
- ICTP-IAEA Workshop, Trieste (Italy), May 2023 - Oral presentation
- CIBiosDA-1, Abidjan (Côte d'Ivoire), November 2022 - Oral presentation

- COVID-19 & Climate Symposium, Online, August 2020 - Poster presentation
- 100th AMS Meeting, Boston (USA), January 2020 - Oral presentation
- 99th AMS Meeting, Phoenix (USA), January 2019 - Oral presentation
- Global Climate Variability Symposium, Ankara (Turkey), April 2019 - Oral presentation
- S2D Forecast Conference, Boulder (USA), September 2018 - Poster presentation
- AGU Fall Meeting, Washington DC (USA), December 2018 - Participation
- CLIVAR PIRATA-PREFACE, Paris (France), December 2016 - Oral presentation
- ANACIM-NOAA Workshop, Dakar (Senegal), July 2016 - Oral presentation
- ESP Young Researchers Day, UCAD, Dakar (Senegal), June 2016 - Oral presentation and coordinator
- 2nd One Medicine One Science Conference, Minnesota (USA), April 2016 - Poster presentation
- EGU General Assembly, Vienna (Austria), April 2016 - Poster presentation
- Science & Innovation Fair, Dakar (Senegal), November 2015 - Poster presentation and coordinator
- CAMES Scientific Days (JSDC-2), Dakar (Senegal), November 2015 - Oral presentation
- Modeling & Climate Variability Colloquium, Liège (Belgium), July 2015 - Poster presentation
- AIC 2014 Conference, Dijon (France), July 2014 - Poster presentation on climate and health

---

GRANTS & FUNDING

- EU FP7 (QWeCI, PREFACE)
- GEF (NAP Support Project)
- USAID / Save the Children / UNICEF
- WHO - IPD Vector Project
- NOAA Postdoc Fellowship (UCAR)
- Erasmus Mundus Doctoral Grant

TECHNICAL SKILLS / COMPUTER SKILLS

- Proficient in **MS-DOS, MS Office** (Word, Excel, PowerPoint), **MATLAB, R, Python, GIS** (Geographic Information Systems), **Linux operating system**, and **FORTRAN** language.

  • Experienced in data analysis of observational and model output datasets.

  • Hands-on experience with impact and forecast models such as:

  - **LMM** (Liverpool Malaria Model)
  - **VECTRI** (Vector-borne disease community model of the ICTP, Trieste)
  - **S4CAST** (Sea Surface Temperature-based Statistical Seasonal Forecast model)
  - **CPT** (Climate Predictability Tool)
  - **CMT** (Climate Monitoring Tool)AI & Machine Learning Expertise

- AI & Machine Learning Expertise

FOREIGN LANGUAGE SKILLS

- **French:** Proficiency in speaking, understanding, and writing.
- **English:** Advanced proficiency in speaking, understanding, and writing.
- **Spanish:** Upper-intermediate proficiency in speaking, understanding, and writing.

REFERENCES

- **Dr. Wassila Mamadou Thiaw**
  NOAA Center for Weather and Climate Prediction
  Climate Prediction Center
  5830 University Research Court
  College Park, Maryland 20740, USA
  ✉ **Email:** wassila.thiaw@noaa.gov

- **Prof. Amadou Thierno Gaye**
  Cheikh Anta Diop University
  High Polytechnic School
  Laboratory of Physics of Atmosphere and Ocean - Simeon Fongang (LPAO-SF)
  BP: 5085 Dakar-Fann
  ✉ **Email:** atgaye@ucad.edu.sn

- **Minister of Health Ibrahima Sy**
  Ministry of Health and Social Action, Fann résidence
  Rue Aimé Césaire, BP 4024 Dakar, Senegal
  ✉ **Email:** ibrahima.sy@cse.sn